



CHEAPSKATE'S NUMERAI ML INFRASTRUCTURE

BUILDING TOTALLY FREE COMPUTING INFRASTRUCTURE FOR MODEL TRAINING AND AUTOMATED
SUBMISSION OF PREDICTIONS.

JOSEF „THE FOSSIL“ ŠVENDA

CHEAPSKATE
=
DRŽGREŠLE

WELCOME IN CZECH REPUBLIC

- CZECH LANGUAGE:
STRČ PRST SKRZ KRK! = STICK FINGER THRU THROAT!
- CZECH BEER
- NICE TO LIVE
- NON RELIGIOUS
- FUNNY



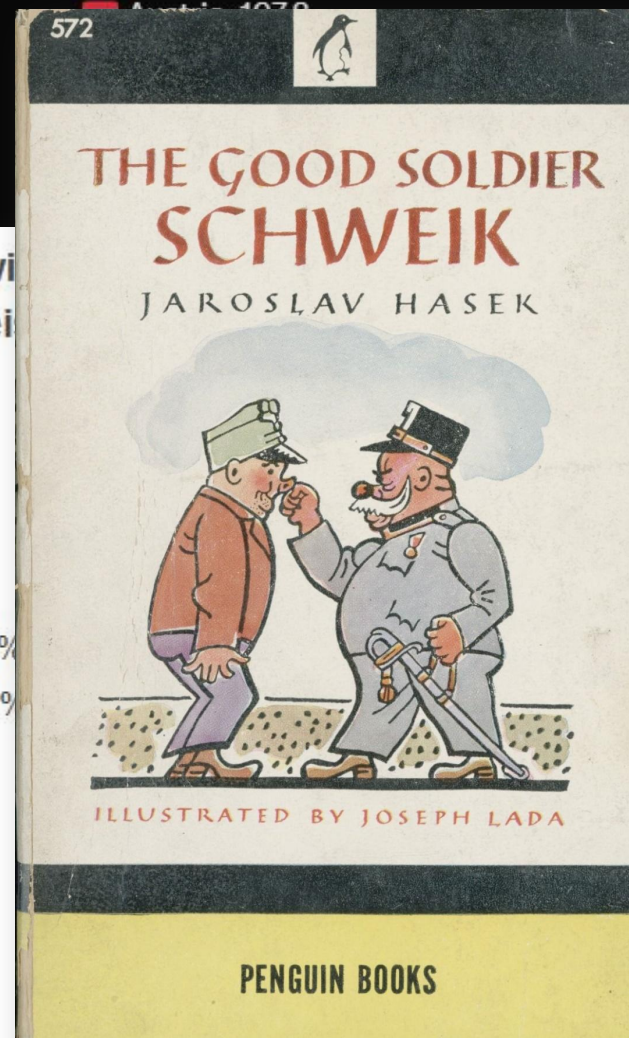
World of Statistics @stats_feed · Jun 21
Beer consumption per capita (liters per year):

Czechia: 140.0

Austria: 107.0

Top 10 Countries with Percentage of Atheists 2017:

1. China - 91%
2. Japan - 87%
3. Sweden - 78%
4. Czech Republic - 76%
5. United Kingdom - 73%
6. Belgium - 72%
7. Estonia - 72%
8. Norway - 70%
9. Australia - 69%
10. Denmark - 68%



Countries in the World
Peace Index — lower is

7
- 1.269
8
96
0
01
16
— 1.318
326
1.357
9

NUMERAI

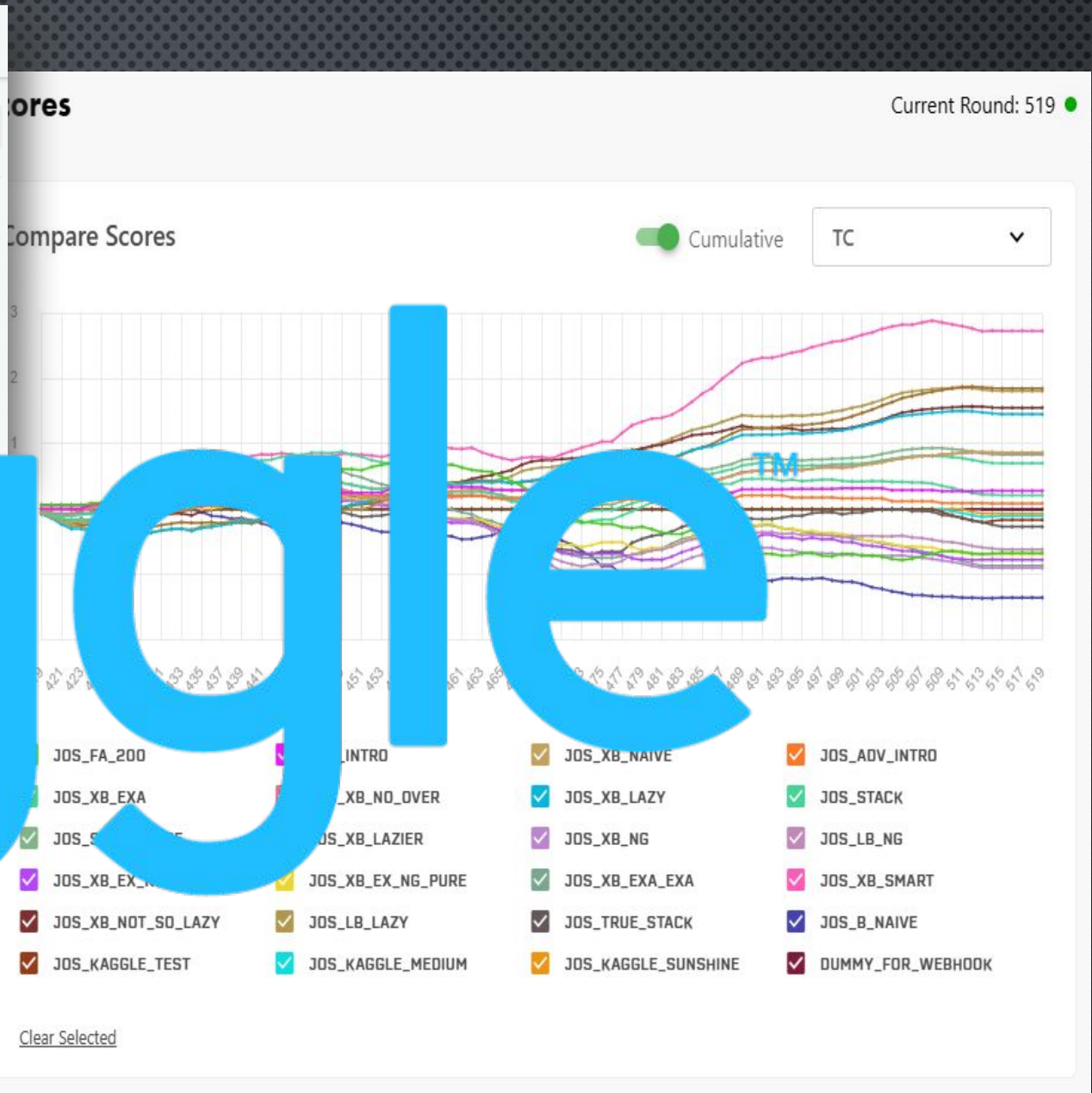
svendaj

Expand

STATS

136 days visited 22h time 5h recent read time 161 topics viewed 1.7k posts read

31 ❤️ given 63 ❤️ rec 1 bookmark 3 topics created 28 posts created

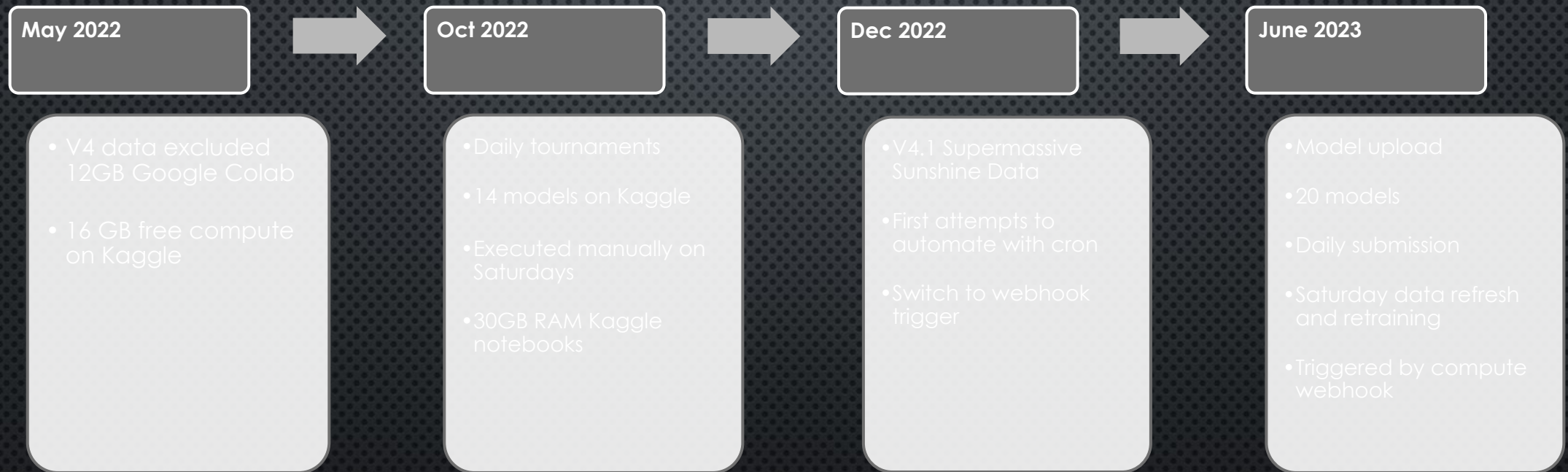


kaggle

\$0

COSTS OF COMPUTE

HOW IT HAPPENED...



LIMITS OF FREE

WHAT YOU CAN DO IN KAGGLE

Limits

4 CPUs, 30GB RAM

GPU, 2 CPUs, 13GB RAM

TPU, 4 CPUs, 16GB RAM

12 hours CPU&GPU, 9h TPU

20GB disk space

Max 10 notebooks concurrently running

Chaining of notebooks

Kaggle API

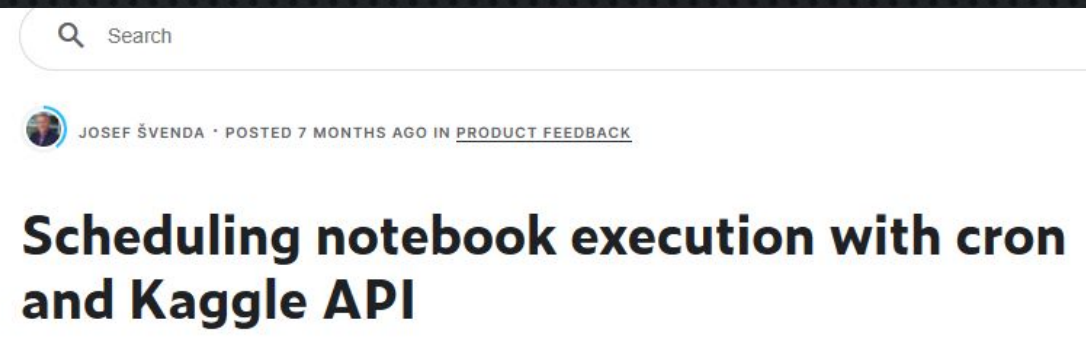
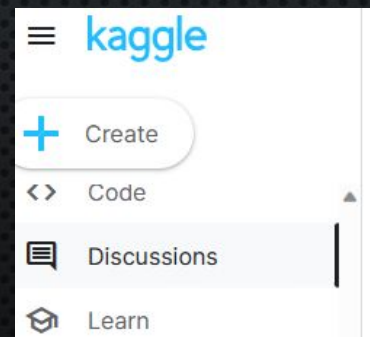
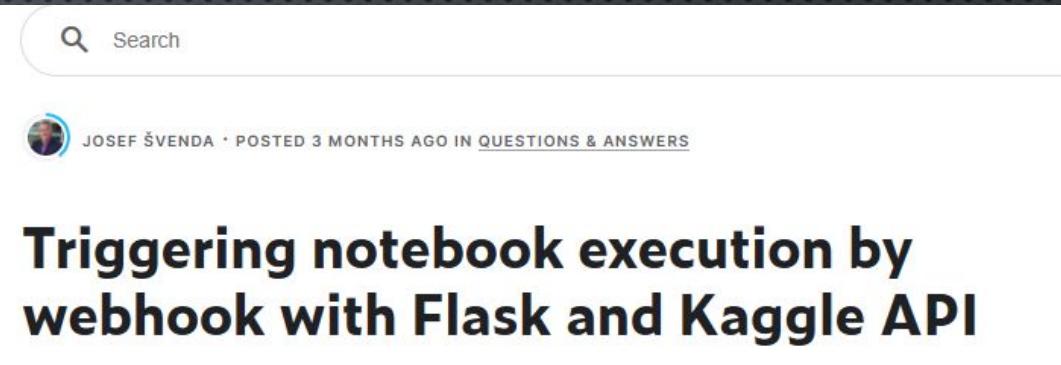
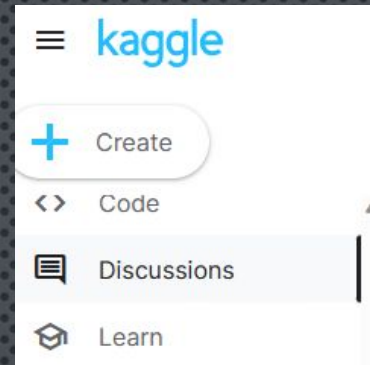
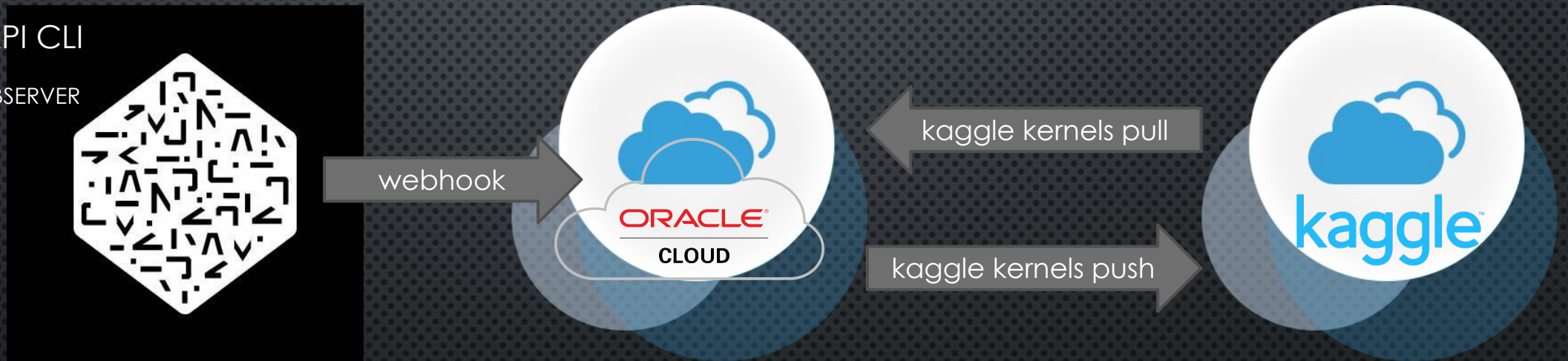
- running Kaggle notebooks from outside

Enablers



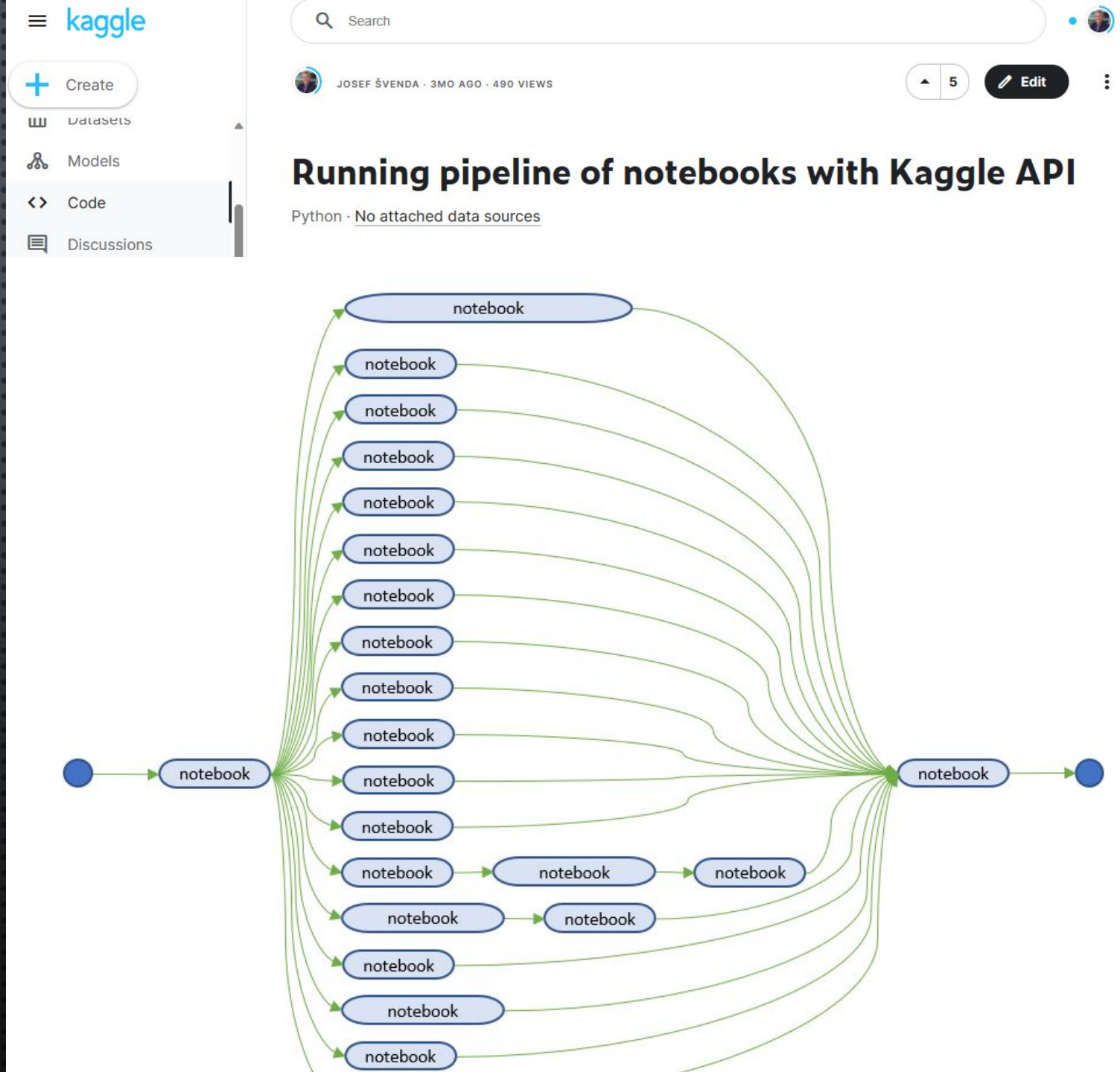
TRIGGERING RUN ON KAGGLE

- ORACLE CLOUD, „ALWAYS FREE“, UBUNTU VM, 1 CPU WITH BOOT DISK
- PYTHON
- KAGGLE API CLI
- FLASK WEBSERVER



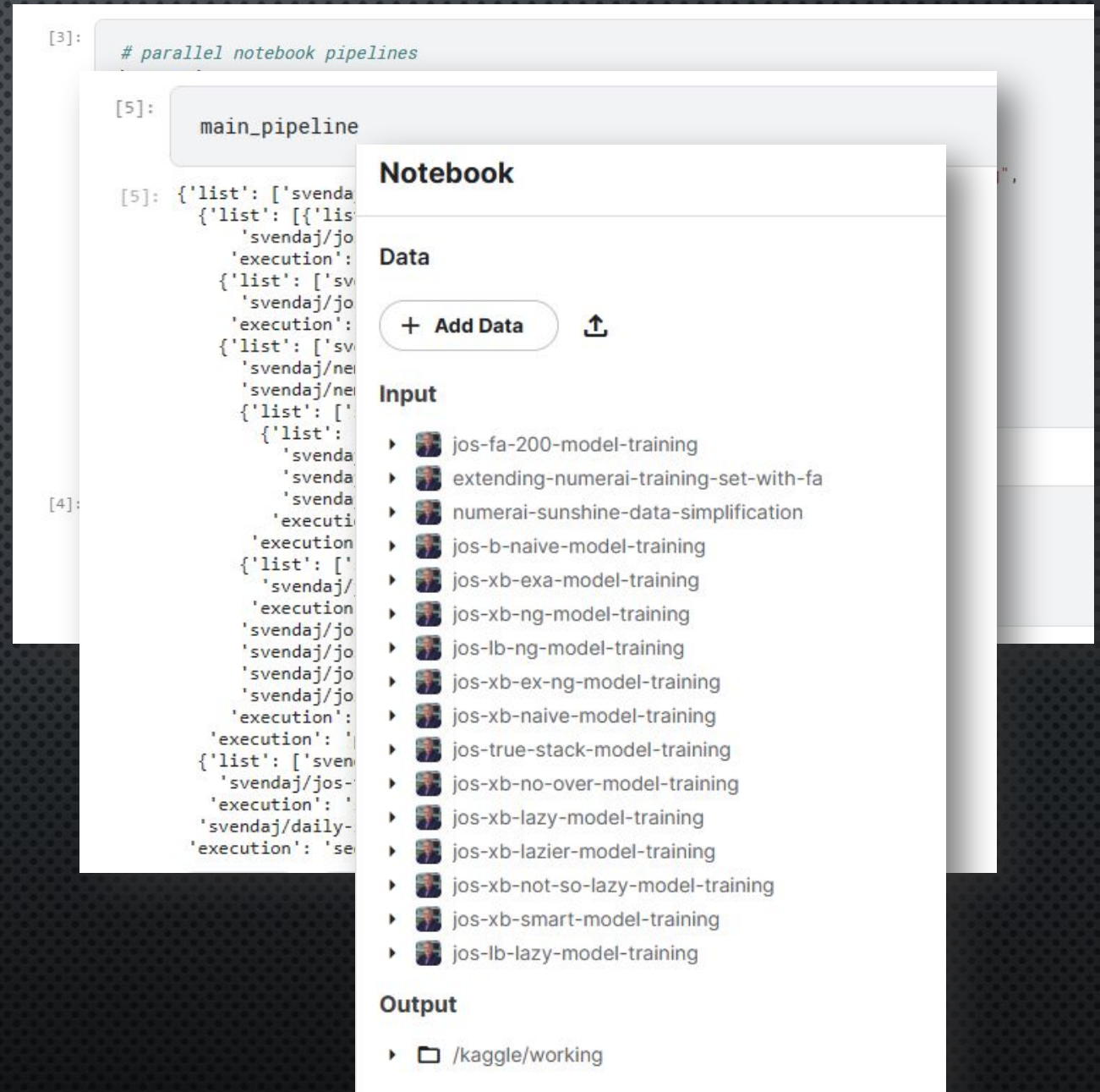
KAGGLE NOTEBOOK ORCHESTRATOR

- RUNNING OTHER NOTEBOOKS
 - SEQUENTIALLY
 - PARALLEL
- MUST FINISH IN 12 HOURS
- KAGGLE API CLI
- CONTROLLING PYTHON DICT



MY KAGGLE INFRASTRUCTURE (BEFORE MODEL UPLOAD)

- WEEKLY RUN TRIGGERED AT SATURDAY ROUND OPENING
- SINGLE ORCHESTRATING NOTEBOOK:
 - DOWNLOADS [WEEKLY_NUMERAI_DATASET](#) AND MAKES IT PUBLIC
 - LAUNCHES 3 PARALLEL PIPELINES WITH 20 NOTEBOOKS & FOLLOWS WITH STACKING PIPELINE OF 2 NOTEBOOKS
- DAILY SUBMISSION NOTEBOOK
 - INPUT ALL TRAINED MODELS PICKLED
 - DOWNLOADS LIVE_INT8 DATA
 - PREDICTS AND SUBMITS



KAGGLE BONUS

- LEARNING — COURSES
- COMMUNITY
- COMPETITIONS
- WARNING „KAGGLE IS ADDICTIVE“

Featured Code Competition

JPX Tokyo Stock Exchange Prediction

Explore the Tokyo market with your data science skills

 Japan Exchange Group · 2,033 teams · 9 months ago

OverviewDataCodeDiscussionLeaderboardRulesTeam

SubmissionsLate Submission

Leaderboard

Search leaderboard

PublicPrivate

The private leaderboard is calculated with approximately 87% of the test data. This competition has completed.

Prize Winners

#	Δ
1	—
2	—
3	—

Featured Code Competition

AMP®-Parkinson's Disease Progression Prediction

Use protein and peptide data measurements from Parkinson's Disease patients to predict progression of the disease.

 AMP®-PD · 1,805 teams · 2 months ago

OverviewDataCodeDiscussionLeaderboardRulesTeam

SubmissionsLate Submission

Leaderboard

Search leaderboard

PublicPrivate

The private leaderboard is calculated with approximately 87% of the test data. This competition has completed. This leaderboard reflects the final standings.

Prize Winners

#	Δ	Team	Members	Score	Entries	Last	Solution
1	▲ 1	Connecting dots...		60.042	242	2mo	
2	▲ 4	no luck all skill		60.077	244	2mo	
3	▲ 2	Hajime Tamura		60.141	305	2mo	
4	▼ 3	Chris Deotte		60.171	213	2mo	

BEST THINGS IN LIFE ARE FREE!!!

... BUT GOOD FINANCIAL REWARD IS ALWAYS APPRECIATED ;-)